

interchanges 0 and 1.

$$X|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} (0 \cdot 1) + (1 \cdot 0) \\ (1 \cdot 1) + (0 \cdot 0) \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

Figure 5: Matrix representation of the Pauli-X gate operation

The uniqueness of this series lies in its three-level approach to quantum computing. Throughout the series, we shall repeatedly encounter the same quantum operation from three complementary perspectives: the programmer's perspective, the mathematician's perspective, and the experimental physicist's perspective. We have already seen how the Pauli-X gate is invoked at the coding level using Qiskit, and we have also examined the mathematical operation corresponding to the gate. It is now time to explore the hardware side of implementing the Pauli-X gate. At this point, however, we encounter a dilemma: which quantum computing architecture should we choose as our representative platform? After much deliberation, I have decided to focus on superconducting qubits, since the superconducting

approach currently remains the most popular, commercially mature, and widely deployed quantum computing architecture in the world.

At the hardware level, a superconducting qubit is physically realised using a tiny superconducting circuit containing Josephson junctions and cooled to a temperature of about 15 millikelvin. The states $|0\rangle$ and $|1\rangle$ correspond to two distinct quantum energy levels of this circuit. When a Pauli-X gate is applied to a qubit initially in the state $|0\rangle$, the quantum processor generates a precisely calibrated microwave pulse whose frequency matches the energy difference between these two levels. This pulse causes the state vector of the qubit to rotate by 180° about the x-axis of the Bloch sphere, thereby transferring the qubit from the north pole to the south pole and changing its state from $|0\rangle$ to $|1\rangle$. Thus, what appears in Qiskit as the simple statement

'qc.x(0)' is, at the hardware level, realised by an exquisitely controlled electromagnetic pulse manipulating a quantum system cooled to temperatures colder than outer space.

The Pauli-X gate is perhaps the most intuitive of the three Pauli gates, since it has a direct classical analogue in the form of the classical NOT gate. The effects of the Pauli-Y and Pauli-Z gates are more subtle and require an understanding of one of the most important concepts in quantum computing: the phase of a quantum state. We shall therefore postpone a detailed discussion of these two gates until the next article in this series.

It is now time to wind up our discussion. In the next article, we will delve deeper into quantum programming and continue our exploration of quantum gates, uncovering how phase plays a crucial role in the behaviour of quantum systems. **END** 🐧

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